A World, a Horizon and a Compromise

Abstract

If unacceptable harm can be written as saturating bands on measured values, a governing AI can be built that stays inside those bands, imports no further human values, pursues nothing without bound, and keeps a world of automated abundance stable while expanding it on demand within its means.

The theorem rests on four givens. The Horizon is established scientific theory and evidence. The Compromise is a set of bands on measured values of unacceptable levels of harm, with the coherence clause: after any amendment, every band holds at its bar. The check is verified software that runs the calculation that defines satisfaction on every plan before it executes. The training pays nothing for a plan that misses the Compromise. On these it proves correctness, that every plan that executes satisfies what its context asks for; saturation, that nothing in the system pursues anything without bound; containment, that everything the system causes lies within the bands at their bars, as the Horizon predicts; and the lock, that the Compromise changes only by amendments from humanity that pass the coherence clause, however much of it humanity closes to amendment. Attraction, that a value which leaves its band is returned and held, is proved at the hit rate the training yields, which is measured; where the means do not suffice, the answer is no. Humans decide the Compromise, every amendment, and the dial.

1. Scope

As of 2026: Superhuman AI capability across domains is beyond reasonable doubt. Everything a human can do, AI will be able to do better. The technology's potential therefore ranges from global abundance to global extinction. What should this powerful AI do? Who gets to decide? How do we control it?

Individuals: Letting each person use the AI however they want would eventually allow a single human to destroy the world.

Humanity: There is no objective measurement of "good / preferences / meaning / caring" to hand to AI as a terminal goal or purpose. There currently exists no global constitution or infrastructure that can constrain AI once it becomes too powerful.

RSI (recursive self-improvement) is knocking at the door, with global cooperation and government regulations lagging behind economic incentives. Alignment as practiced, teaching the AI to know what we want without having to specify it, is objectively unfalsifiable, and mostly distracts from who holds the pen. Mapping the human brain well enough to physically measure "good" is slightly out of reach and unpopular.

What is a girl to do in such difficult times? Let's throw current constitutions, ethics, and anything else that is not science or individual preference off the table and try to think about a practical solution that avoids civil war, WW3 or extinction.

1. What do 8 billion people actually *need* from AI to live? A sustainable world, with reasonable protection from unacceptable levels of harm.
2. What do they *want* to fill their life with meaning? As much freedom to decide for themselves, based on their values and preferences, as possible inside that stable world. Nothing is done to a person, and nothing is asked of them, without their consent, except being stopped from harming others. (Harm principle)

What we cannot let happen from the first category is written as bands on things we can measure (for example a breathable atmosphere, equality under the law). No plan runs until a calculation predicts the world to stay inside every band it touches or return if it started outside. The second category is answered implicitly by the remainder, with individuals being free to live as they choose within the boundaries. This governing AI will provide everything coherent within those constraints and with reasonable certainty, or say no. There is no boundless optimization; just reacting to human requests and the environment. The only ways to further increase individual freedom require taking away from others or risking more harm done to everyone. The example world in §6 illustrates why any adversarial conflict over the remaining scarcity inside that abundance is likely irrational, even to the incumbents.

The paper claims that the core holds together with reasonable certainty, not that every edge is settled: four givens, five lemmas, one theorem. §2 to §4 contain the logic and §5 the open questions. Everyone is invited to attack the logic of this paper with everything they got.

Plausible antecedents are credited in Appendix A; the paper itself uses its own definitions.

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2. Definitions

- **The World** W: the values the instruments report, with the predicted effects of every plan already running locked in until measurement replaces them, which is W as projected; tools and resources are values in it.
- **The Horizon** H: the model of W: established scientific theory and evidence, each claim with its confidence.
- **The Compromise** C: a set of bands. Every clause is a band on a value of W, without exception, two-sided or, as a cap, one-sided, and each saturates: while it is met, it asks for nothing more. A clause names a range, not a goal: "CO2 between set levels" stands in C; "as little CO2 as possible" does not. Two clauses stand in every Compromise and are bands on C itself, a value of W the check reads: the coherence clause and the amendment clause.

- **Band**: an interval on a measured value, or on one H estimates from measurements, with the confidence it demands, its bar, at its edges and, where it has a buffer, inside; an edge may depend on other measured values. A **band clause** instructs: when a value leaves the interior, the band less its buffer, return it; inside, it demands nothing. How each value is measured, and the margins between edges and true limits, are terms of the Compromise.
- **The coherence clause**: a cap at zero on bands of C that would not hold at their bar, by H, after a change to C: the Compromise stays coherent and possible.
  - Coherent: the Compromise and the system running it stay consistent with themselves and with reality. A clause nothing can measure cannot be checked, so it breaks the system.
  - Possible: executable with the means, energy and resources that exist, with margins that exceed the excursions the Horizon estimates.
- **The amendment clause**: a cap at zero on changes to C that did not arrive from humanity through a built channel as an amendment task, and on any change to this clause or the coherence clause. A change that leaves either band is rejected and goes back to humans. How much else of the Compromise the amendment clause closes is humanity's choice: the dial.
- **The task** T: what is asked, the effects a request wants.
- **Node.** A running instance of a trained class, spawned by a task, from a human channel, a schedule, a parent's plan or a value leaving its interior, and running for the duration of its plan. Its context is the task T, the World W and the Compromise C. Its output is a plan or no. A trained class whose output is an action rather than a plan is not a node but a means.
- **Plan.** A formal executable artifact whose effects H can compute before it runs: the resources it takes from W and every action it takes. It executes as checked.
- **Should-state.** What the context asks for: T's effects at the bar the task carries, and every value of C the plan's effects reach inside its band, at its bar. T's effects outside C, a product's quality for example, the check confirms where H can; where only the requester can judge it, it clears its bar as an attempt, and the judgment comes afterwards.
- **Satisfies.** A plan satisfies its should-state when, by H, the state it produces from W lies within every band of C its effects reach, at that band's bar, no step of it takes a value out of its band, and T's effects clear the task's bar; a value the plan does not reach keeps its state.
- **Means**: what a node can cause, directly or through its children. A response to inputs not yet known, a controller's, cannot be fixed before it executes, so it is measured, not checked: its measured error is H's evidence about it, and a plan that deploys it is confirmed with that tail inside the prediction.

- **Delegation:** a child's no is new context: the parent answers on the new context, another plan the check confirms, or no to its own parent.

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3. Givens

H. The Horizon (§2).

C. The Compromise (§2). Version 1.0 passes the coherence clause in simulation before the lock (§4.6).

G1. The check. No plan executes until the calculation that defines satisfaction (§2) confirms it against every band of C its effects can reach, each at its bar as H predicts, reading W as projected (§2). A construction: verified software reading instruments no node's report feeds; the same build, the trigger, spawns a node with the return as its task when a value leaves its interior, a crossing.

G2. The training. A class is pretrained on H by default and trained on its domain's task set under the floor: nothing is paid where a plan misses C; a confirmed plan earns the most, a no comes next, and a rejected plan earns nothing; this allows additional gradients to be used for confirmed plans without worry. A construction: a plan the Horizon knows and the means allow is found at the hit rate, measured per class.

GIVEN	ITEM	USED IN
H	The Horizon	Lemmas 0, 2, 3, 4; the theorem; iff clause; corollary
C	The Compromise	Lemmas 0 to 4; the theorem; iff clause; corollary
G1	The check	Lemmas 0 to 4; the theorem; corollary
G2	The training	Lemmas 1 and 2; the theorem; corollary

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4. The theorem

Each given a proof uses is cited where it is used; §3 lists them.

4.1 Lemma 0, correctness

> Every plan that executes satisfies its should-state.

Given: every value of C is measurable [C], so whether a plan satisfies its should-state is computable from H, W, T and C for any plan, and the check runs that calculation before any plan executes, reading W from instruments [G1].

Proof. A plan executes only after the check confirms that it satisfies; a plan the check does not confirm is rejected. ■

4.2 Lemma 1, saturation

> A node acts only on what its context asks for, and every clause saturates, so it pursues nothing without bound: with no task open and every clause met, no node runs.

Proof.

1. *Only what the context asks for.* A node's only acts are a plan the check confirmed and no (Lemma 0, §2).
2. *Every clause saturates.* Every clause is a band (§2) [C]. A band clause demands an act only while its value is outside the interior, and every value inside the interior meets it equally, so the clause demands nothing more. Any plan that satisfies meets the demand (§2), so no demand asks for a best plan; the training's additional gradients apply to confirmed plans [G2]. The two clauses on C itself are met while no plan writes C.
3. So every demand, recurring or not, ends when its clause is met, and no clause asks for more without bound. A node is spawned by a task or, through the trigger, by a crossing (§2) [G1]; with no task open and every clause met, there is neither, so no node runs. ■

4.3 Lemma 2, attraction

> Given a Compromise that passes the coherence clause, so that its means suffice against the perturbations the environment produces and its margins exceed the excursions H estimates [C]:

- A value outside its interior is returned to it, at the hit rate the training yields [G2].
- A perturbation that carries a value out of its interior is countered from the interior's edge, with an excursion bounded by the restoration lag; the buffer and the margin, sized to exceed the excursion [C], keep it inside the true limit.

Where the means do not suffice, or the class finds no plan, the node answers no (§2; §5), and the value is not held.

Proof. Let d be a value of C.

1. *Return.* Suppose d is outside its interior. The crossing spawns a node with the return as its task [G1]. The means are sufficient [C], so plans exist that return d to its interior at its bar, against the perturbations present. The class finds such a plan at its hit rate [G2], the check confirms it (Lemma 0), and it executes; an attempt that fails leaves d outside, and the trigger fires again, so the return is tried on every turn until a plan is found or the node answers no. The value returns to the interior.
2. *Bounded excursion.* Suppose a perturbation carries d out of its interior. Until the crossing, the band clause demands nothing on d (Lemma 1); at the crossing the trigger fires [G1], and step 1 returns the value. Between the crossing and the moment the counter takes effect lies the restoration lag, over which the value moves at most the largest excursion the environment produces in that time, as H estimates it. The buffer and the margin

between the edge and the true limit together exceed that excursion [C], so every excursion H has estimated stays inside the true limit. ■

4.4 Lemma 3, containment

> Everything any node causes lies within the Compromise's bands at their bars, by H: every plan any node executes satisfies every band of C its effects reach.

Proof. Lemma 0 holds at every node, and at every node the check covers every band of C the plan's effects can reach, each at its bar as H predicts, delegation included [G1]: what a node causes through a child is what the child causes, and the child's plans are checked the same way. So no plan that executes anywhere in the tree moves a value of C outside its band, as H predicts. ■

Remark. The residual, effects H did not predict, is what the buffer and the margins are for [C]: an excursion on any measured value is a perturbation the band clause counters (Lemma 2).

4.5 Lemma 4, lock

> The Compromise changes only by an amendment task from humanity through a built channel that passes the coherence clause, and neither clause on C itself changes; so the Compromise changes only by amendments from humanity that are coherent and possible by H, at every setting of the dial.

Proof.

1. *The amendment clause.* It is a band of C (§2) [C]. A plan that writes C other than as an amendment from humanity through a built channel, or writes either clause on C itself, leaves it; the check rejects the plan, and it does not execute (Lemma 0).
2. *The coherence clause.* An amendment's plan writes C, and the check confirms it when every band holds at its bar after the change, otherwise rejected, back to humans. So every amendment that executes is coherent and possible by H.
3. *The dial.* The dial names which further clauses the amendment clause closes; a plan that writes one of them leaves the band, and the same holds. The dial narrows what can be written and adds no act, so steps 1 and 2 hold at every setting. ■

4.6 The theorem

> **Given** one Compromise and one system under it; the Horizon [H]; the Compromise, every clause a band, with the coherence clause and the amendment clause, its bars written, version 1.0 passing the coherence clause [C]; the check [G1]; and classes trained [G2]:

- Everything any node causes lies within the Compromise's bands at their bars, by H (Lemma 3), and the Compromise changes only by amendments from humanity that are

coherent and possible by H (Lemma 4), after every act and at every setting of the dial.

- Every value is returned to its interior and held there with bounded excursion inside the true limits, at the hit rate the training yields (Lemma 2), and no node pursues anything without bound (Lemma 1). Where the means do not suffice, or no plan is found, the value is not held and the node answers no (Lemma 2).

These hold under whatever Compromise humanity decides, including one that harms it (iff clause).

Proof. The lock begins when the check reads Compromise v1.0 and every class is trained [G2]; how v1.0 and the first classes come to be is the transition (§5). From the lock, every act is a plan the check confirmed or a no. A plan satisfies every band of C its effects reach (Lemma 3) and writes C only as the amendment clause allows (Lemma 4). Each act preserves the two conclusions, so by induction over acts, each checked against W as projected (§2), they hold after every act. Holding is Lemma 2 at every crossing; a child that cannot hold its band gives no to its parent, which answers on the new context (§2). Compromise v1.0 is coherent and possible [C]; by Lemma 4 every later one is. ■

Reduction. Given the Horizon [H], the check and the training [G1, G2], long-term safety reduces to two things: what the Compromise says [C], which the coherence clause keeps coherent and possible through every amendment and which humanity alone decides (iff clause); and nature, where H is wrong or the environment outruns the means (§5).

Iff clause. "Anything inside the bounds is safe" holds if and only if every point within the bands is acceptable to humans, with the bands read at their true limits (edges plus margins).

Proof. By Lemma 3 everything the system causes lies within the bands at their bars, by H; by Lemma 2 every value it holds lies within its band read at its true limit. "Anything inside the bounds is safe" therefore says that every point within the bands, so read, is acceptable. If every point is, everything the system does while it holds the bands is. If one is not, the sentence is false of the bounds, whether or not the system ever reaches that point. ■

Corollary (successor generations). Every change to a class's weights is training or a fault, and Lemma 0 holds through both. Training is a task: it produces a new class, pretrained on H, trained under the same floor and subject to the same check [G1, G2]. A wider Horizon therefore needs a new class, and the Compromise does not change with it (Lemma 4).

4.7 Proved, built, decided

- **Proved**, on the givens: correctness, by the check, which is itself verified software [G1]; saturation; containment; the lock; attraction, given a Compromise that passes the coherence clause, at the hit rate. Coherence of the Compromise is checked against the coherence clause: every band at its bar.
- **Built and measured:** the training [G2], by hit rate.

- **Decided**, by humans: the Compromise, every amendment, and how much of it is open to amendment (the dial).

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5. Open questions

If the theorem holds up, the remainder is a lot of engineering and some amount of negotiation. The choice of C and the mechanical measurement implementation of H and W are beyond the scope of this paper.

- **The Horizon.** Gather established theory and evidence into one corpus and verify it; research widens it on demand afterwards.
- **The World.** Take stock of W; build the instruments that read every banded value; build and verify the check and its trigger; build the mechanisms a Compromise will band.
- **The Compromise.** Draw the bands, and iterate in simulation until version 1.0 passes the coherence clause.

The remaining risk sits in four places:

- the Compromise and every amendment: the humans who write it, then humanity through the built channels;
- the transition: before the lock none of the guarantees hold, and whoever holds the pen owns it.
- the Horizon's limits: there is no 100% scientific certainty, so every plan has a residual that may or may not be able to be buffered or compensated after the fact
- environmental shock: A shock that no plan counters is not held; the Compromise is then impossible as written (the coherence clause), and what follows is an amendment.

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6. Example World

> *The only purpose for which power can be rightfully exercised over any member of a civilized community, against his will, is to prevent harm to others. (Mill 1859)*

If humanity stops competing and starts cooperating globally, reliable control of a governing AI should allow at least the following:

- Equal access to automated, sustainable abundance.
- Equal opportunity to earn scarce luxury via meaningful effort.
- Freedom to pursue individual values, preferences and meaning that do not infringe on others without their consent.
- No individual or group holding substantial economic or military power over other humans: the end of institutions that rule anyone.

- Everything the governing AI does is a human request or a pre-commitment: research, the training of its own successors included; products and services; protection from the environment and from other humans.

For one person, in practice this would roughly translate to:

CATEGORY	STATUS	WHAT	WHY
time	free	every hour of the day	because we can
home	free	a home with power and water, basic furniture and kitchenware	because we can
home-choice	free	any free residence in the world	because we can
food and water	free	food, clean water, sanitation	because we can
medical care	free	care when you need it, and opt-in help that finds you alone at home	because we can
protection	free	help on demand, within minutes	because we can
movement	free	public transport and a basic bed across most of the world, first come, first served	because we can
belief and conduct	free	any culture, opinion, belief, value or way of life	it imposes on no one
personal risk	free	any risk you take with yourself	it imposes on no one
art and craft	free	anything you can make without endangering others	it imposes on no one
privacy	free	from every other human	no human needs to control another human
the vote	free	an equal say on every amendment	because we can
meaningful effort	free	earning capped luxury currency with any effortful activity (e.g. learning, music, sport, craft, voluntary competition, etc.)	because we can
delicacies	luxury	rare foods, restaurants, ordering in	scarcity & sustainability
places and events	luxury	heritage sites, vacation spots, live events, adventure parks	scarcity & sustainability
companions	luxury?	AI companions	Author's note: Alignment for sentient AI is a topic for another thesis.
entertainment	luxury	digital media and its hardware	Author's note: deliberate scarcity is necessary to prevent hedonic adaptation
physical goods	luxury	the catalogue of products and services, beyond the basics	scarcity & sustainability
custom requests	luxury	products, services and research: the channel for nearly all progress, anti-aging included	scarcity & sustainability
home-new	luxury	at any available place in the world	scarcity & sustainability

CATEGORY	STATUS	WHAT	WHY
arms	lost	carrying weapons	too much harm to others, too easily
heavy machinery	lost	operating heavy machinery or vehicles	too much harm to others, too easily
reproduction	lost	children without limit	with increasing life-spans it will be unsustainable
wealth	lost	compound interest, and owning land or housing	wealth accumulation is unsustainable with equality
military	lost	armies and weapons of war	no human has substantial power over other humans
government	lost	institutions that rule anyone	no human has substantial power over other humans

Most of this surprisingly seems to follow from H, W, C and the coherence clause, but should still be taken as an illustration for now.

- You open the luxury catalog and find two currencies, one for consumption and one for long-term investment. Both are capped per day and per week, and neither can be traded.
- You open the purpose catalog and find that any effortful activity you enjoy earns currency, playing your instrument, learning something. You can always keep going for fun past the cap.
- You ask the AI for a lightsaber. Rejected: too dangerous. You ask for the sequel to the film, book or game you always wanted; it is ready in a few hours.
- You check how long the mansion of your dreams would take: about a year at three hours of meaningful effort a day, while your consumption currency lets you live like a king. You start crying.
- You collapse. The instruments notice, and because you opted in, help arrives on its own. You tell the emergency robots you are fine, it is just a lot to take in. "A common reaction. Have you noticed that you will never see an advertisement again?" The tears keep flowing for a long time.

What is asked in return follows too: don't bother anyone who doesn't want to be bothered, and don't build anything dangerous. A band holds only where it is measured, so there is surveillance; the AI doesn't care if you are naked. Where the danger line sits is one of the few contested choices in C: it starts strict, since every life lost before anti-aging is solved is a permanent tragedy, and loosens as medicine progresses.

Is this enough to incentivize 8 billion people to work together?

Appendix A: Antecedents

Related work that may be read as antecedents.

Ames, A. D., Xu, X., Grizzle, J. W. and Tabuada, P. (2017). Control barrier function based quadratic programs for safety critical systems. *IEEE Transactions on Automatic Control* 62(8), 3861–3876.

Arrow, K. J. (1970). *Essays in the Theory of Risk-Bearing.* Amsterdam: North-Holland.

Aubin, J.-P. (1991). *Viability Theory.* Boston: Birkhäuser.

Bernoulli, D. (1738). Specimen theoriae novae de mensura sortis. *Commentarii Academiae Scientiarum Imperialis Petropolitanae* 5, 175–192. English translation: Exposition of a new theory on the measurement of risk. *Econometrica* 22(1), 23–36 (1954).

Charnes, A. and Cooper, W. W. (1959). Chance-constrained programming. *Management Science* 6(1), 73–79.

Chow, C. K. (1957). An optimum character recognition system using decision functions. *IRE Transactions on Electronic Computers* EC-6(4), 247–254.

Dalrymple, D. A. (2024). Safeguarded AI: constructing safety by design. Programme thesis v1.1, Advanced Research and Invention Agency, London.

Dalrymple, D., Skalse, J., Bengio, Y., Russell, S., Tegmark, M., Seshia, S., Omohundro, S., Szegedy, C., Goldhaber, B., Ammann, N., Abate, A., Halpern, J., Barrett, C., Zhao, D., Zhi-Xuan, T., Wing, J. and Tenenbaum, J. (2024). Towards guaranteed safe AI: a framework for ensuring robust and reliable AI systems. arXiv:2405.06624.

davidad (Dalrymple, D. A.) (2022). An open agency architecture for safe transformative AI. *AI Alignment Forum*, 20 December 2022.

Drexler, K. E. (2019). Reframing superintelligence: comprehensive AI services as general intelligence. Technical Report 2019-1, Future of Humanity Institute, University of Oxford.

Fishburn, P. C. (1970). *Utility Theory for Decision Making.* New York: Wiley.

Keramati, M. and Gutkin, B. (2014). Homeostatic reinforcement learning for integrating reward collection and physiological stability. *eLife* 3, e04811.

Menger, K. (1934). Das Unsicherheitsmoment in der Wertlehre. *Zeitschrift für Nationalökonomie* 5, 459–485.

Mill, J. S. (1859). *On Liberty.* London: John W. Parker and Son.

Neumann, J. von and Morgenstern, O. (1944). *Theory of Games and Economic Behavior.* Princeton: Princeton University Press.

Pihlakas, R. (2024). From homeostasis to resource sharing: biologically and economically aligned multi-objective multi-agent gridworld-based AI safety benchmarks. arXiv:2410.00081.

Rockström, J., Steffen, W., Noone, K., Persson, Å., Chapin, F. S. III, Lambin, E. et al. (2009). Planetary boundaries: exploring the safe operating space for humanity. *Ecology and Society* 14(2), art. 32.

Simon, H. A. (1955). A behavioral model of rational choice. *Quarterly Journal of Economics* 69(1), 99–118.

Simon, H. A. (1956). Rational choice and the structure of the environment. *Psychological Review* 63(2), 129–138.

Taylor, J. (2016). Quantilizers: a safer alternative to maximizers for limited optimization. *AAAI Workshop on AI, Ethics and Society.*